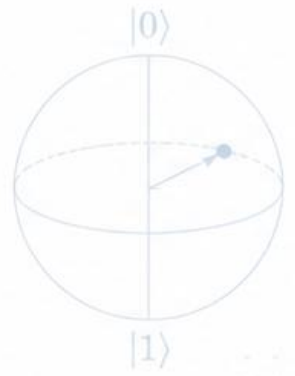


Introduction to Quantum Computing

CEN 1417 – Quantum Computing

Outline

- **Simple Qubit Models**
 - Understanding qubits using a two-level atom model.
- **Wave-Particle Duality and Double-Slit Experiment**
 - Explaining superposition and quantum interference.
- **Classical Bit vs Qubit**
 - Comparing classical bits with qubits and their general state.
- **Dirac Notation and Probability Amplitudes**
 - Understanding $|\psi\rangle$, $|0\rangle$, $|1\rangle$, α , and β .
- **Basis States and Superposition**
 - Exploring $|0\rangle$, $|1\rangle$, and superposition states.
- **Quantum Measurement**
 - Showing why measurement gives one random result and collapses the state.
- **Vector Notation**
 - Representing qubit states as vectors to prepare for quantum gates

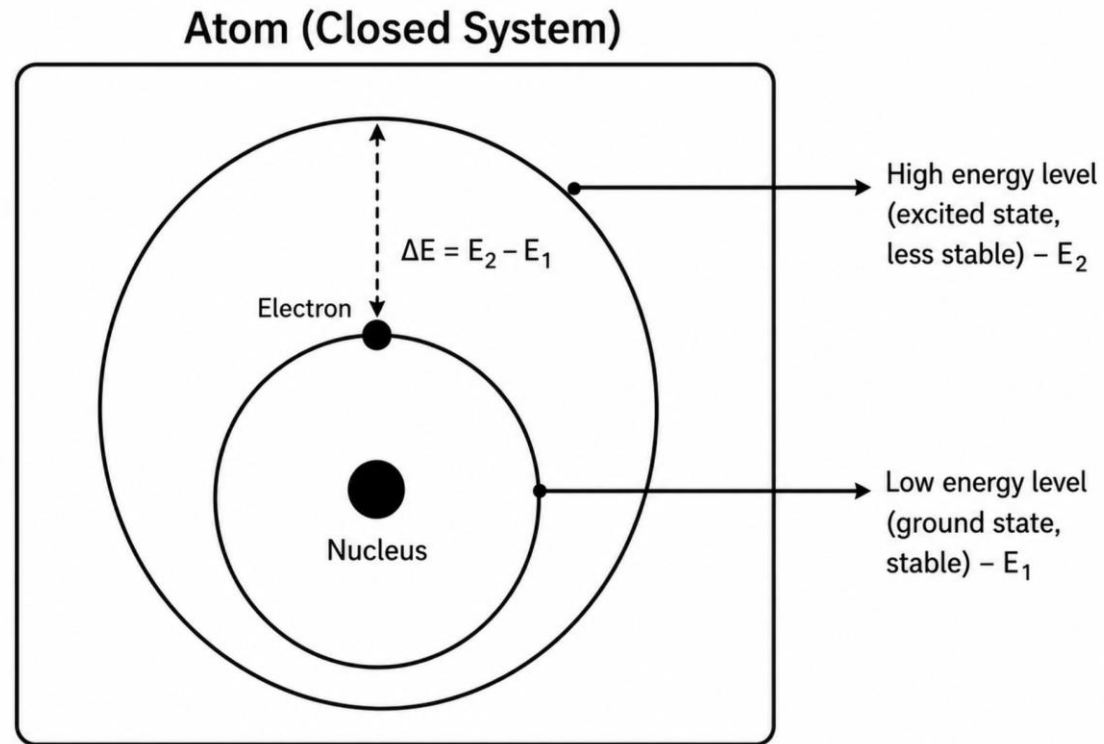


Two-Level Atom as a Simple Qubit Model -Bohr-like Model

 **1. Classical-like mapping:**
 $|0\rangle$ and $|1\rangle$ can be represented by two distinct energy states.

 **2. Superposition:**
 $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

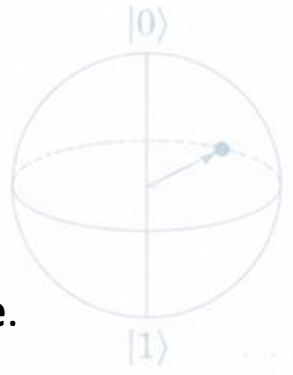
 **3. Measurement**
collapses the state to either $|0\rangle$ or $|1\rangle$.



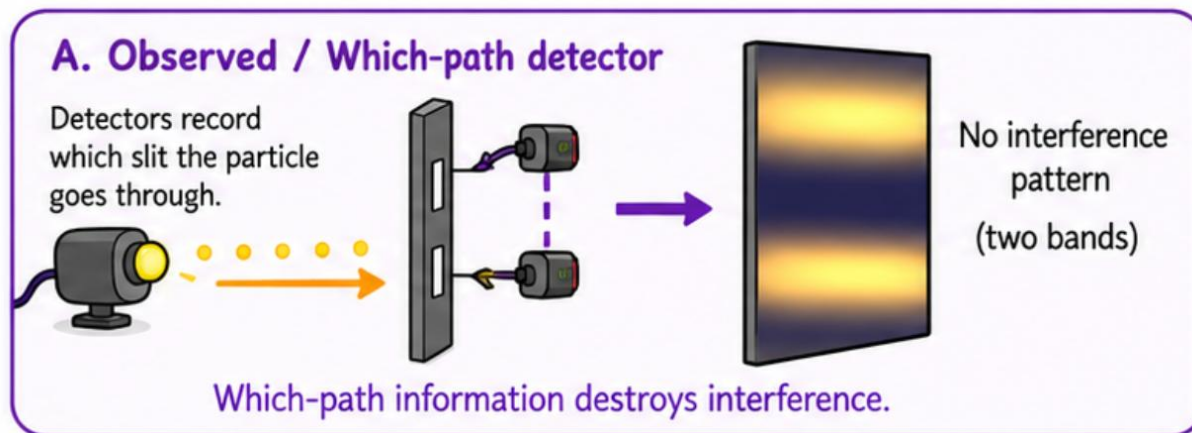
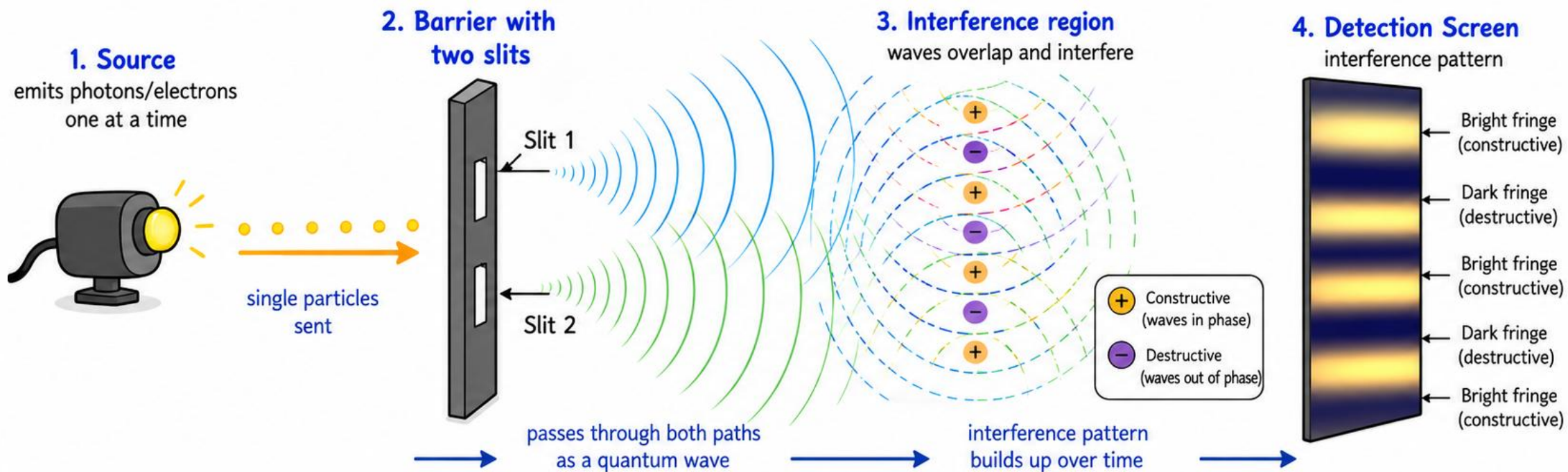
- This is a simplified conceptual model of a qubit using a two-level quantum system, not a literal representation of all quantum computers.

The Double-Slit Experiment

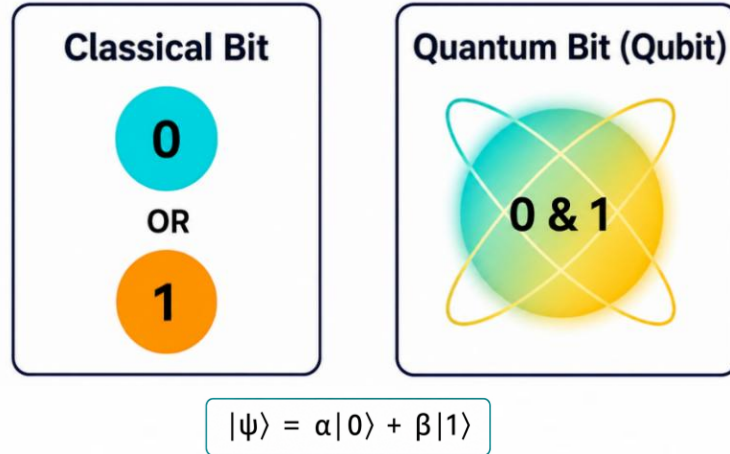
- The double-slit experiment shows that quantum particles, such as photons or electrons, can exhibit both wave-like and particle-like behavior. When these particles pass through two slits, they produce an interference pattern, demonstrating a key quantum phenomenon known as **quantum interference**. Together, **wave-particle duality** and **quantum interference** are fundamental **ideas in quantum mechanics**.
 - It is called wave-particle duality because quantum objects, such as photons or electrons, can show **wave-like behavior**, such as **interference**, and **particle-like behavior**, such as being detected at a specific point on the screen.
- **Superposition:** Before measurement, a quantum particle can be described as being in a superposition of two possible paths, passing through the left slit and the right slit at the same time.
- **Quantum Interference:** Quantum interference occurs when the probability amplitudes of these two possible paths combine, producing the interference pattern observed on the screen.
- In **quantum computing**, **superposition** allows qubits to represent many possible states at once, while **quantum interference** is used by quantum algorithms to amplify the probability of correct answers and reduce the probability of wrong answers before measurement.



The Double-Slit Experiment

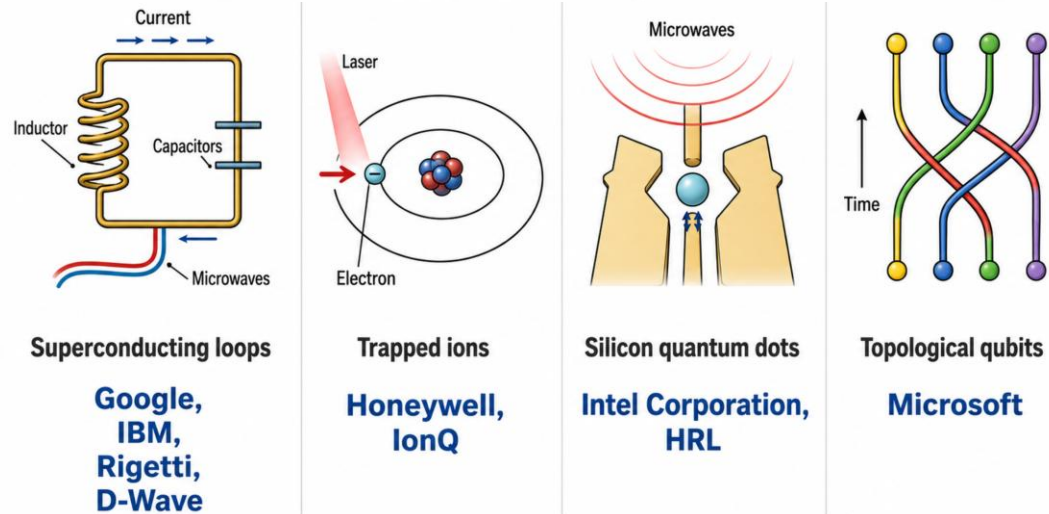


Classical Bit vs Quantum Bit (Qubit)



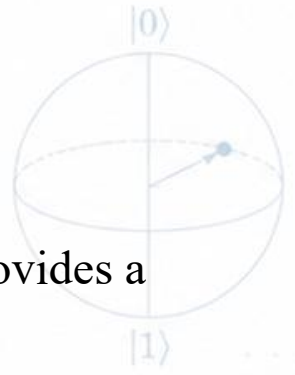
- A classical **bit** is represented by ordinary electrical states (1 or 0) in transistor-based circuits, while a qubit is represented by **quantum states of a physical system**.
- Qubit is the basic unit of quantum information.
- In quantum computing, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ represents the general state of a single qubit, where α and β are complex probability amplitudes.
- The values of α and β determine the probabilities of measuring $|0\rangle$ or $|1\rangle$. For example, a qubit may have: **50% probability of being measured as $|0\rangle$ and 50% as $|1\rangle$** or **93% probability of being measured as $|0\rangle$ and 7% as $|1\rangle$**
 - The total probability must always add up to 100%.
- A **qubit** does not give both results at the same time when it is measured. When we measure a qubit, we get only one result: either **$|0\rangle$ or $|1\rangle$** . However, before measurement, the qubit can exist in a **superposition** of both states.

Physical Implementations of Qubits



- A **qubit** is not made from one specific material. It is a unit of quantum information that can be physically implemented using different **controllable quantum systems** (physical system that behaves according to the laws of quantum mechanics), such as **superconducting circuits**, **trapped ions**, **photons**, or **electron**. In each implementation, **two** controllable quantum states are selected to represent $|0\rangle$ and $|1\rangle$, while also allowing **superposition** before measurement.
- **Note:** At this level, we are not focusing on the detailed physics or hardware design. The key idea is that a qubit can be implemented using different controllable quantum systems.
- In a **qubit**, we choose two quantum states from a physical system to represent $|0\rangle$ and $|1\rangle$. These states are **controllable**, meaning we can **prepare**, **manipulate**, and **measure** them

Dirac notation or bra-ket Notation



- **Dirac notation** is a standard way to write quantum states in quantum computing. It is important because it provides a simple, compact way to represent quantum states, making it easier to describe qubits and superpositions.
- The state of a single **Qubit** before **measurement** is usually written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

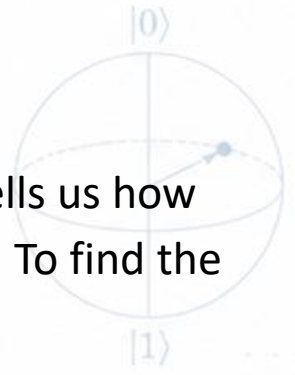
- This **equation** is written using **Dirac notation**. It represents the state of a single qubit **before** measurement. The equation means that a qubit can be described as a combination of the two basic quantum states: $|0\rangle$ and $|1\rangle$. Unlike a classical bit, which must be either 0 or 1, a qubit can be in a **superposition** of both states before measurement. This means that both $|0\rangle$ and $|1\rangle$ can have non-zero probability amplitudes. After measurement, the qubit usually no longer remains in this general **superposition** form. Instead, it **collapses** to one definite outcome: either 0 or 1.

Dirac Notation II: Meaning of the Symbols



- The symbol ψ is called **Psi**, represents the general state of one qubit before **measurement**, which can be a superposition of the basis states $|0\rangle$ and $|1\rangle$.
- The notation $|\rangle$ is called **ket notation**. It is a mathematical notation used to represent **quantum states** in quantum mechanics. When a state label is placed inside it, such as $|\psi\rangle$ It represents a quantum state.
- α (Alpha) represents the probability amplitude of the basis state $|0\rangle$. It tells us how much the qubit is associated with $|0\rangle$ before measurement.
- β (Beta) represents the probability amplitude of the basis state $|1\rangle$. It tells us how much the qubit is associated with $|1\rangle$ before measurement.

Dirac Notation III: Probability Amplitudes vs Probabilities



- The symbol α (**Alpha**), represents the **complex probability amplitude** of the qubit being in the state $|0\rangle$. It tells us how strongly the qubit is connected to the $|0\rangle$ state before measurement. However, α is not the probability itself. To find the probability of measuring the qubit as 0, we calculate $|\alpha|^2$.
- The symbol β (**Beta**), Beta represents the **complex probability amplitude** of the qubit being in the state $|1\rangle$. It tells us how strongly the qubit is connected to the $|1\rangle$ state before measurement. Like α , β is not the probability directly. To find the probability of measuring the qubit as 1, we calculate $|\beta|^2$.
- For a valid single-qubit state, the probability amplitudes must satisfy the **normalization condition**:

$$|\alpha|^2 + |\beta|^2 = 1$$

In the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, α and β are called **complex probability amplitudes** because they can be complex numbers. They are not probabilities directly. The actual probabilities are obtained by taking the absolute value squared:

$$P(0) = |\alpha|^2 \text{ and } P(1) = |\beta|^2$$

Qubit Basis States: $|0\rangle$ and $|1\rangle$

- **Case 1: The Qubit is in State $|0\rangle$**

The state of the qubit is:

$$|\psi\rangle = 1|0\rangle + 0|1\rangle$$

This means the qubit is completely in state $|0\rangle$.

Measurement probabilities: $P(0) = 100\%$ and $P(1) = 0\%$

Therefore, this is not a superposition state, because the qubit has no part in state $|1\rangle$.

- **Case 2: The Qubit is in State $|1\rangle$**

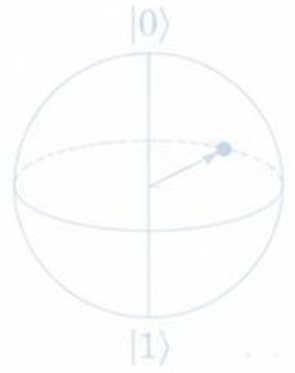
The state of the qubit is:

$$|\psi\rangle = 0|0\rangle + 1|1\rangle$$

This means the qubit is completely in state $|1\rangle$.

Measurement probabilities: $P(0) = 0\%$ and $P(1) = 100\%$

Therefore, this is not a superposition state, because the qubit has no part in state $|0\rangle$.



Qubit States: Superposition State



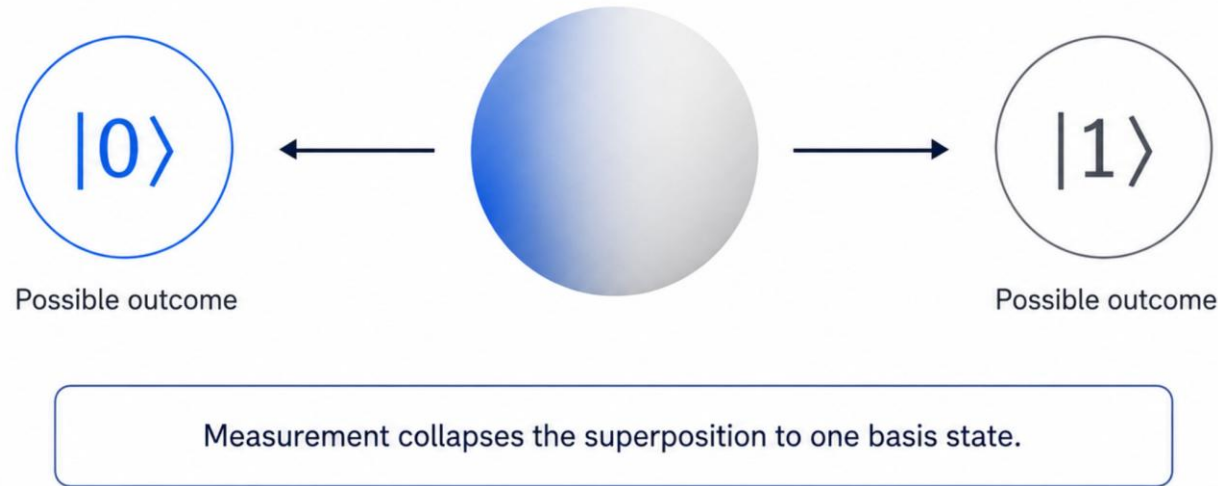
- **Case 3: The Qubit is in a Superposition State**

An example of a superposition state is:

$$|\psi\rangle = (1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$$

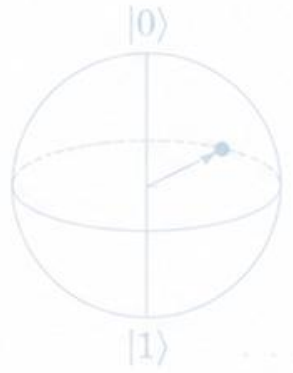
- This means the qubit is not only in state $|0\rangle$, and not only in state $|1\rangle$. Instead, it is described as a combination of both states before measurement.
- Measurement probabilities: $P(0) = 50\%$ and $P(1) = 50\%$. Therefore, this is a superposition state because both $|0\rangle$ and $|1\rangle$ have **non-zero** amplitudes.

Superposition in Quantum Computing



- **Superposition** allows qubits to be in **one quantum state** that contains **amplitudes** for multiple possible basis states **before** measurement. This enables quantum parallelism, where one quantum operation can affect all components of the superposition simultaneously.
- We cannot directly read all these components after measurement; measurement gives only one outcome. Therefore, the real power of quantum computing does not come from superposition alone. It comes from using **interference**, and often **entanglement**, to increase the probability of measuring the correct answer and reduce the probability of incorrect answers.

Quantum Reality vs Measurement



- **Superposition Cannot Be Seen Directly**

- Before measurement, a qubit can be in a superposition, meaning it has probability amplitudes for both $|0\rangle$ and $|1\rangle$. However, we cannot directly “see” or measure the superposition itself. When we measure the qubit, the device does not show us the full superposition, such as “75% $|0\rangle$ and 25% $|1\rangle$.” Instead, it gives only one result $|0\rangle$ or $|1\rangle$.

Measurement Gives One Random Result



- **Measurement Gives One Random Result Based on Probabilities**

- Before measurement, we can prepare a qubit in a specific quantum state. This process is called **state preparation**. For example, the qubit may be prepared so that it has a 75% probability of being measured as $|0\rangle$ and a 25% probability of being measured as $|1\rangle$. A single measurement will still give only one result $|0\rangle$ or $|1\rangle$. The result is random, but it follows the probabilities of the prepared quantum state. For example, if we repeat the same preparation and measurement 1000 times, we may get results close to:

753 times $|0\rangle$ **and** 247 times $|1\rangle$

- From these repeated measurements, we can estimate that the prepared quantum state had approximately 75% probability for $|0\rangle$ and 25% probability for $|1\rangle$.

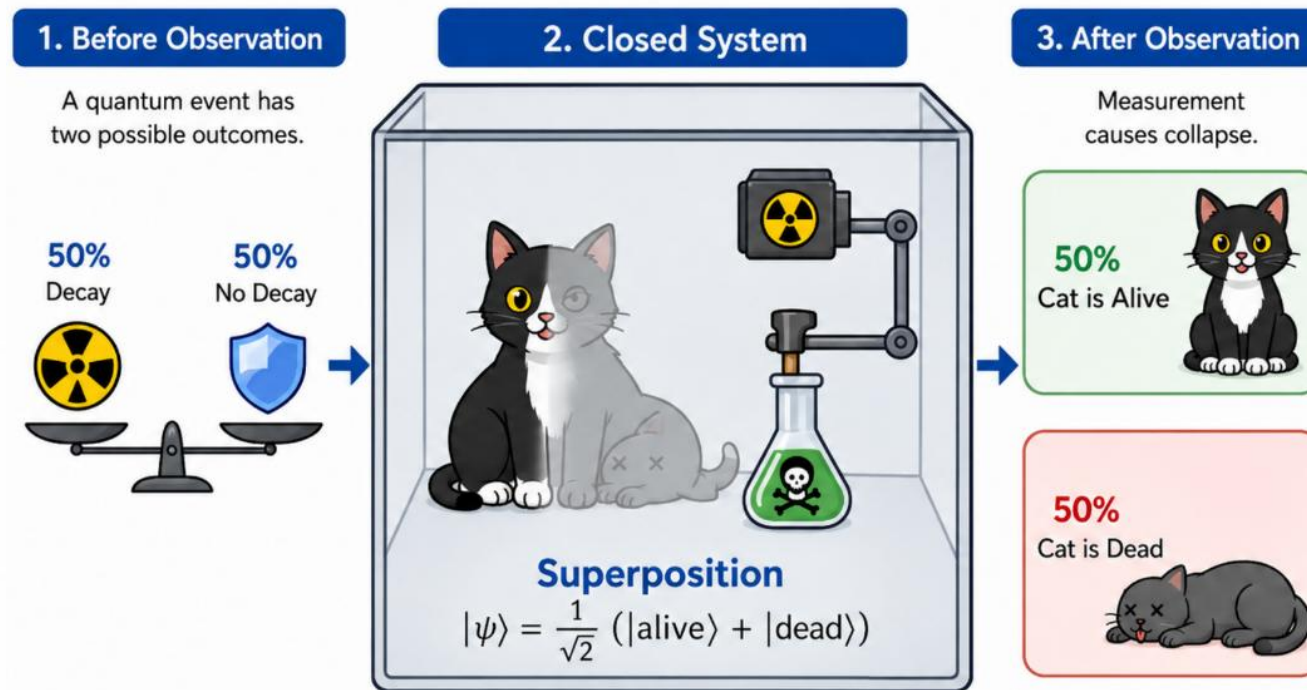
Repeated Measurements Reveal Probabilities



- **Measurement Changes the Qubit State**

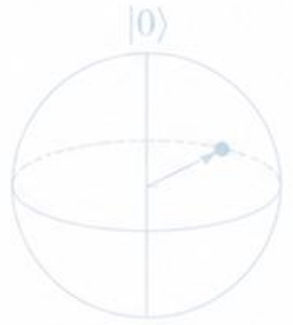
- Before measurement, the qubit is prepared in a superposition state. After measurement, the qubit collapses (**Quantum State Collapses**) into the measured basis state. For example, if the measurement result is $|0\rangle$ then the qubit becomes $|0\rangle$ after the measurement. This means we cannot measure the same qubit many times to discover its original superposition, because the first measurement changes the state.
- To estimate the original probabilities, we must repeat the whole process many times:
 - **Prepare the qubit → Create the superposition → Measure the qubit**
- Then we use the statistical distribution of the results to understand the probabilities.

Schrödinger's Cat: Superposition and Measurement



- Proposed by **Erwin Schrödinger** in 1935, this thought experiment illustrates how a quantum system can exist in a superposition of possible states before measurement, while observation reveals only one definite outcome.

Vector Notation for a Qubit State



- Vector notation is a mathematical way to represent a qubit state as a column of numbers. Each number in the vector represents the probability amplitude associated with one basis state. For a single qubit, the two basis states are $|0\rangle$ and $|1\rangle$. Therefore, in **Dirac notation** a general qubit state can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- The same state can be represented using **vector notation** as:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Basis States in Vector Notation

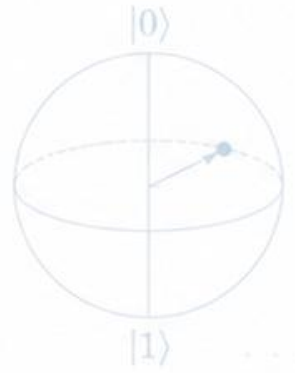


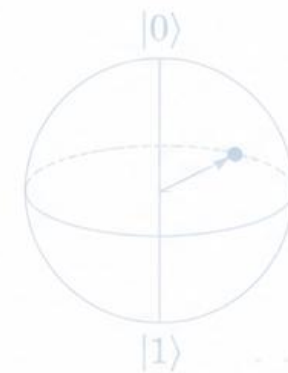
- A qubit in superposition is described by a single quantum state vector that contains probability amplitudes for the possible measurement outcomes, such as $|0\rangle$ and $|1\rangle$. This does not mean that the qubit has multiple quantum states at the same moment
- Examples of Basis States in Vector Notation:
 - If the qubit state is $|\psi\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ then this means $|\psi\rangle = 1|0\rangle + 0|1\rangle$ therefore, the qubit is in the state $|0\rangle$ with 100% probability, and measuring it will always give 0.
 - If the qubit state is $|\psi\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ then this means $|\psi\rangle = 0|0\rangle + 1|1\rangle$ therefore, the qubit is in the state $|1\rangle$ with 100% probability, and measuring it will always give 1.
- At any specific moment, a single qubit has one state vector. This vector contains the amplitudes α and β . Different values of α and β can represent different quantum states, but they must satisfy $|\alpha|^2 + |\beta|^2 = 1$.

Why Vector Notation Matters

Vector notation is important because it gives us the mathematical form needed to perform quantum computations.

- Vector is used to represent a quantum state because a qubit can be in a superposition of the basis states $|0\rangle$ and $|1\rangle$. Each basis state has its own probability amplitude. These amplitudes are represented together in a vector.
- In quantum computing, a qubit state is represented as a vector, a **quantum gate** is represented as a **matrix**, and **applying a gate** means multiplying the matrix by the state vector:
 - Qubit state = Vector
 - Quantum gate = Matrix
 - Applying a gate = Matrix $\times$ Vector
 - New state = Gate matrix $\times$ Current state vector





Thank you for your attention.